

■■ INTELLIGENT PHISHING URL DETECTION

Explainable AI (XAI) Security Analysis Report • Generated on 2026-08-18 08:11:02 UTC

TARGET URL: https://github.com/torvalds/linux	PREDICTION: LEGITIMATE
Domain: github.com	Phishing Probability: 0.0% (Safe Risk)
Model Architecture: Random Forest	Model Confidence: 100.0%

1. Explainable AI (SHAP Feature Attribution)

The chart below lists the top features that contributed most heavily towards or against the phishing classification:

Feature Name	Extracted Value	Contribution	Impact Direction
URL Length	33	-0.101	-0.101 (Safe/Benign)
Prefix/Suffix Dash	0	-0.067	-0.067 (Safe/Benign)
Suspicious Keywords Count	0	-0.064	-0.064 (Safe/Benign)
TLD Risk Factor	0.1	-0.062	-0.062 (Safe/Benign)
Domain Length	10	-0.039	-0.039 (Safe/Benign)
Shannon Entropy	4.2	-0.028	-0.028 (Safe/Benign)

2. Extracted URL Lexical & Heuristic Features

Feature Metric	Value	Feature Metric	Value
URL Length	33	Subdomain Count	0
Domain Length	10	Dot Count	1
HTTPS Encryption	Yes	Hyphen Count	0
Direct IP Host	No (Domain Name)	Digit Count	0
Suspicious Keywords	0	Shannon Entropy	4.20
Contains @ Symbol	No	TLD Risk Rating	10%

3. AI Cybersecurity Mitigation Recommendations

- Safe to browse, but always verify the exact browser address bar before entering sensitive credentials.
- Ensure multi-factor authentication (MFA) is enabled on your accounts for layered protection.